

notions of interpersonal connections.

Hyper-real platforms create new avenues for businesses, enabling virtual commerce, advertising, and brand experiences. The emergence of virtual economies and digital assets within the metaverse presents economic opportunities for content creators and entrepreneurs. Collaboration among hyper-real platform developers is crucial to establish standards, interoperability, and shared experiences.

Open source initiatives and partnerships can drive innovation and accelerate the development of the metaverse. As the metaverse continues to develop, hyper-real platforms will shape the way we interact, work, play, and experience the digital world.

Use cases of hyper-real platforms in metaverse

From gaming and entertainment to architecture, education, retail, healthcare, and beyond, the transformative use cases of hyper-real platforms are vast and diverse (Table 1). As technology advances and these platforms continue to evolve, we can anticipate even more groundbreaking applications that will shape the future of the metaverse, unlocking new levels of realism and interaction that were once unimaginable. Some of the popular use cases where hyper-real platforms can play a vital role are listed below.

Gaming and entertainment.

Hyper-real platforms have the potential to revolutionise the gaming

and entertainment industry within the metaverse. By leveraging advanced graphics rendering, realistic physics simulations, and immersive audio, these platforms can create breathtaking virtual worlds and experiences. Players can step into hyper-realistic environments, interact with lifelike characters, and engage in highly immersive gameplay. From virtual reality (VR) gaming to augmented reality (AR) experiences, hyper-real platforms offer unprecedented levels of immersion, enhancing entertainment experiences and captivating audiences like never before.

Industrial architecture and design. In the realm of architecture and design (e.g., automobile industrial design), hyper-real platforms can reshape the way professionals visualise and present their ideas. Architects and designers can cre-

ate virtual models of buildings and spaces, offering clients an immersive walkthrough experience that closely resembles reality.

Hyper-real platforms enable detailed material simulations, accurate lighting effects, and realistic spatial interactions, empowering stakeholders to make informed decisions during the design process. This immersive approach not only enhances collaboration but also saves time and resources by identifying potential issues before construction begins.

Education and training. Hyper-real platforms hold significant promise for education and training

applications within the metaverse. By creating realistic simulations of various scenarios, such as medical procedures, hazardous environments, or historical events, these platforms can provide immersive and experiential learning opportunities. Students and professionals can engage in hands-on training, practice skills in a safe environment, and gain practical knowledge that translates into real-world scenarios. The hyper-realistic nature of these platforms enhances engagement and knowledge retention, transforming the way we learn and train across diverse fields.

Retail and e-commerce. Hyper-real platforms offer exciting opportunities for the retail and e-commerce sectors in the metaverse. Virtual stores can replicate the physical shopping experience by providing immersive 3D environments where

TABLE 2
ECONOMIC OPPORTUNITIES AND BUSINESS MODELS OF HYPER-REAL PLATFORMS

Economic Opportunities	Business Models
Virtual real estate and property development	Virtual land acquisition and development
	Virtual property rental and sales
	Virtual construction and design services
Digital goods and virtual asset marketplace	Creation and sale of virtual products
	Virtual asset trading and auctions
	Limited edition virtual collectibles
Virtual events and experiences	Virtual event organisation and ticket sales
	Sponsored virtual experiences and exhibitions
	Virtual event merchandise and partnerships
Advertising and brand integration	Immersive virtual advertising campaigns
	Branded virtual environments and experiences
	Virtual product placements and sponsorships
Virtual services and consultation	Virtual consulting and expert services
	Virtual training and mentorship programs
	Personalised virtual services and advice
Platform development and infrastructure	Development of hyper-real platforms
	Infrastructure and maintenance services
	Tools and asset creation for platforms